

OpenADMET PXR Structure Corrected Submission Report

Objective

This report documents the corrected full submission package prepared after external blind-test validation identified a subset of failing structures in the original baseline archive. The correction workflow was restricted to the failing compounds and used two targeted remediation strategies: protonation-state repair for chemically invalid cationic states, and GNINA docking fallback for ligands whose UniDock-Pro poses remained highly strained after minimisation.

Corrected Submission Package

- corrected archive members: 184
- corrected archive size (bytes): 8354265
- total blinded ligands in submission: 184
- corrected ligands replaced in the full archive: 25
- charge-state corrected ligands: 11
- GNINA-fallback corrected ligands: 14

Root Cause Analysis Of The Failing Subset

Two failure modes were identified among the 25 rejected structures:

1. Chemically invalid protonation states assigned to amide- or lactam-like nitrogens.
2. Catastrophically strained docked poses from UniDock-Pro hybrid docking, especially for ring-rich ligands.

The first class was corrected by rejecting protonated states on nitrogens directly attached to carbonyl-like or sulfonyl-like groups. The second class was corrected by widening the UniDock-Pro pose harvest, applying explicit MMFF94 strain filtering, and invoking GNINA docking when all UniDock-Pro poses for a ligand-template pair remained above the strain threshold.

Protonation-State Corrections

For the following ligands, the prepared docking state was reverted from an invalid cationic amide/lactam state to the chemically reasonable neutral input state:

- x00644-1
- x00773-1
- x02782-1
- x02909-1
- x02914-1
- x03260-1
- x03279-1
- x03331-1
- x03387-1

- x03432-1
- x03463-1

Charge-state remediation summary:

- corrected ligands with changed prepared state: 11
- corrected baseline exports written: 11

Strain-Driven GNINA Fallback

For the remaining neutral failures, UniDock-Pro hybrid docking was retained as the first docking engine, but each ligand-template pose set was screened by MMFF94 strain after GNINA minimisation. When all UniDock-Pro poses remained above the strain threshold, GNINA full docking was run against the same receptor/template, followed by GNINA minimisation and CNN rescoring. The lower-strain engine was then retained for baseline pose selection.

Neutral ligands corrected by GNINA fallback:

- x01016-1
- x02715-1
- x02776-1
- x02797-1
- x02828-1
- x03037-1
- x03096-1
- x03152-1
- x03223-1
- x03234-1
- x03282-1
- x03319-1
- x03400-1
- x03401-1

GNINA fallback summary:

- ligands with fallback-selected pose rows: 14
- fallback pose rows recorded: 140
- corrected neutral baseline exports written: 14

Example GNINA-fallback selections:

- x01016-1 -> template 3R8D, pose x01016-1_3R8D_gninafb_pose_003, CNNaffinity 5.482, minimizedAffinity -5.867
- x02715-1 -> template 1M13, pose x02715-1_1M13_gninafb_pose_001, CNNaffinity 6.911, minimizedAffinity -7.776
- x02776-1 -> template 1NRL, pose x02776-1_1NRL_gninafb_pose_001, CNNaffinity 6.687, minimizedAffinity -9.578
- x02797-1 -> template 1NRL, pose x02797-1_1NRL_gninafb_pose_002, CNNaffinity 7.193, minimizedAffinity -8.989
- x02828-1 -> template 1NRL, pose x02828-1_1NRL_gninafb_pose_002, CNNaffinity 6.384, minimizedAffinity -9.034
- x03037-1 -> template 1NRL, pose x03037-1_1NRL_gninafb_pose_004, CNNaffinity 6.486, minimizedAffinity -8.265
- x03096-1 -> template 1M13, pose x03096-1_1M13_gninafb_pose_005, CNNaffinity 6.910, minimizedAffinity -8.416

- x03152-1 -> template 1NRL, pose x03152-1_1NRL_gninafb_pose_001, CNNAffinity 6.213, minimizedAffinity -9.506

Final Selection And Packaging

The corrected full archive was assembled by overlaying the 25 corrected ligand-protein PDB files onto the original 184-member baseline archive. All corrected PDBs passed the local structural-format checks used in the workflow before packaging.

- replaced PDB entries in the full archive: 25
- retained original baseline entries: 159
- final corrected archive name: submission_pdb.zip

Practical Interpretation

The corrected archive differs from the original baseline submission only for the 25 compounds that failed external validation. Eleven were repaired by chemistry-aware protonation filtering, and fourteen were repaired by switching from strained UniDock-Pro poses to GNINA-docked alternatives selected after minimisation and strain screening. This corrected full archive is therefore the recommended replacement package for resubmission.